WEEKLY LESSON PLAN · 2026-27

Engineering Design & Problem Solving (InSPIRESS)

TEKS §127.785 · Advanced · Senior · Mr. Gavaldon · Da Vinci School

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SEMESTER 1
Week 1

MONDAY Day 1 - Procedures, Syllabus & Design Notebook

Aug 10

OBJECTIVE

Students will set up a professional engineering design notebook to documentation standards (numbered, dated, permanent, witness-ready), sequence the seven stages of the formal engineering design process, distinguish design criteria from constraints, and write a first notebook entry that frames a candidate InSPIRESS payload as a measurable design problem.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - professional standards & the engineering design notebook

ELPS / EB SUPPORT

ELPS/EB supports: pre-teach *criteria*, *constraint*, *iteration*, *specification* with a bilingual visual glossary and icons (PV, DP, VC); model the notebook setup live under the doc cam with numbered steps (CD, RS); provide sentence stems for the framed-problem entry — 'A successful design must ___, measured by ___ (units)'; 'A constraint I cannot violate is ___' (RS); pair EB students to talk through the criteria/constraint sort before writing (PN, WT); allow extra time on the notebook build (ET).

AGENDA · 45 MIN

- **6′** Bell-work: In your notebook, write one sentence on what separates *engineering* from *building*. We'll test it against the professional definition.
- **10′** The engineering design process: seven stages, and why documentation is 'stage zero'
- **10′** The language of a real problem: criteria vs. constraints, framed around the InSPIRESS payload challenge
- **13′** Build your design notebook to professional standard + write your first framed-problem entry
- **6′** Exit ticket: sort four statements into criteria/constraint and write one measurable success criterion

SLIDES / ACTIVITY

Engineering Design & the Professional Notebook

NOTEBOOK

Set it up today: cover, table of contents, numbered/dated pages. First entry = one thing you want to design or present this year.

TURN IN

Design Notebook Standard + Define-Phase Problem Framing

TUESDAY — Day 2 - What Makes a Great Presentation? (+ units)

Aug 11

OBJECTIVE

Students will analyze technical communication using the claim–evidence–interpretation structure, apply significant-figure rules to measured quantities, convert and combine units by dimensional analysis with cancellation shown, and report a measured result with appropriate uncertainty.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - technical communication & presentation; applied measurement (units/dimensional analysis)

ELPS / EB SUPPORT

ELPS/EB supports: the video has a guided viewing sheet with sentence stems and is captioned (RS, CD, VC); pre-teach *significant figures*, *dimensional analysis*, *uncertainty*, *precision/accuracy* with a bilingual glossary and a kept conversion table (PV, DP); the calculation is modeled I-do / we-do / you-do with units written on every line (CD, RS); pair EB students for the you-do (PN, WT); extra time on the problem set (ET).

AGENDA · 45 MIN

- 5′ Bell-work: Write the best presentation you've ever seen and one concrete reason it worked.
- 8′ Watch a working engineer present, then name the *claim → evidence → interpretation* moves she makes
- 7′ The unit of a technical argument: claim → evidence → interpretation
- 13′ Significant figures, units & dimensional analysis — I do / we do / you do
- 8′ Measurement uncertainty: precision vs. accuracy and how to report it
- 4′ Exit ticket

SLIDES / ACTIVITY

Present Like an Engineer: Evidence, Units & Uncertainty

NOTEBOOK

Write the 3 presentation rules. Keep the unit-conversion example with all steps shown.

TURN IN

Technical Communication + Measurement: Evidence, Sig Figs, Units, Uncertainty

WEDNESDAY — Day 3 - Lab & Field Investigation Safety

Aug 12

OBJECTIVE

Students will apply the engineering design process to define a real-world problem and begin designing a solution (Ask, Imagine, Plan).

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - safe, responsible lab & field investigation practices

ELPS / EB SUPPORT

ELPS/EB supports: safety rules shown with icons and a live demo (VC, DP); pre-teach *hazard*, *risk*, *PPE*, *SDS*, *control* with a bilingual glossary (PV); the risk matrix and the mean/range calculation are modeled step-by-step (CD, RS); investigation done in pairs so EB students verbalize procedure and data (PN, WT); extra time to complete the log (ET).

AGENDA · 45 MIN

- 5′ Bell-work: purpose of the engineering design cycle
- 16′ Present: the 5-step engineering design process
- 7′ Pick a real problem (list 3, choose 1)
- 17′ Start the design sheet: Ask -> Imagine (4 idea sketches) -> Plan (Design #1 + materials)

SLIDES / ACTIVITY

Engineering Design Process — Pick Your Problem

NOTEBOOK

Glue in the safety agreement. Record your investigation measurements + what you noticed.

TURN IN

Engineering Design Process sheet - Ask / Imagine / Plan

THURSDAY — Day 4 - Engineering Teams + Your First Mini-Pitch

Aug 13

OBJECTIVE	Students will finish their final design, write a structured AI prompt, and generate + submit a product image.
TEKS	§Eng Design & Presentation TEKS (draft - verify official section) - teamwork & collaboration; short-form presentation
ELPS / EB SUPPORT	ELPS/EB supports: pre-teach *RACI*, *decision matrix*, *weight*, *traceability* with a bilingual glossary and a worked example on the board (PV, DP, VC); the pitch uses a claim–evidence–interpretation sentence-frame planner (CD, RS); teams rehearse together before presenting so EB students practice first (PN, WT); allow notes during the pitch and extra time to plan (ET).
AGENDA · 45 MIN	8′ Bell-work + video (start 1:28): name 2 improvements and why each matters 10′ Model the notebook page (prompt on top, sketch on bottom) + copy the prompt 20′ Finish final design, then generate the product image in Gemini 7′ Submit the image to Google Classroom
SLIDES / ACTIVITY	Finish Your Design + Product Image
NOTEBOOK	Write your team, your role, and your 60-second pitch outline.
TURN IN	AI product image (Gemini) + notebook page (prompt + final design) -> Google Classroom

FRIDAY — Day 5 - Foundations Quiz Game + Accounts

Aug 14

OBJECTIVE	Students will demonstrate mastery of the week's foundations — the design process, criteria vs. constraints, significant figures and unit conversion, measurement uncertainty, risk assessment, and the weighted decision matrix — by solving application problems in a team review, and will create and verify an Autodesk Education account to enable Fusion 360.
TEKS	§Eng Design & Presentation TEKS (draft - verify official section) - review of foundations; digital citizenship & accounts
ELPS / EB SUPPORT	ELPS/EB supports: the review is team-based and low-stakes, with wait time before answers (PN, WT); account steps are numbered and screenshotted in Classroom with a bilingual walkthrough (VC, CD, DP); pre-teach *parametric*, *verify*, *credential* (PV); rephrase problems on request and allow extra time (RS, ET).
AGENDA · 45 MIN	5′ Bell-work: two quick recalls + one calculation (units + safety) 16′ Team review challenge: application problems across the week's skills 16′ Create & verify your Autodesk Education account (Fusion 360) 5′ Account check + info card + digital-citizenship reminder 3′ Exit ticket + preview next week (parametric sketching)
SLIDES / ACTIVITY	Prove It: Week-1 Synthesis + Get CAD-Ready
NOTEBOOK	Note your Autodesk account email (not your password) and one thing you want to model first.
TURN IN	Week 1 Foundations — Cumulative Challenge, Autodesk Account — Info Card + Digital Citizenship